

Unreadable Land: Visual Forensic Investigation of War Simulation

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1. Abstract

In contemporary military gaming, high-fidelity simulations often claim to be “realistic” and “accurate,” while simultaneously shaping public understandings of war. While participants accept the exclusion of unarmed civilians, they also note that the simulation permits the destruction of entire city blocks and villages. This tension, explicitly recognized within the community, points to systematic problems in how violence is represented.

This research employs multi-modal forensic methods to examine how military simulations engineer an appearance of neutrality for global marketability through visual and technical strategies. An essay film alternates rapidly between simulation screenshots and photographic documentation, using variable editing rhythms to deliberately produce perceptual uncertainty. Viewers’ consistent inability to distinguish simulation from reality generates systematically produced experiential evidence that neutrality operates through cultural erasure rather than comprehensive realism. Complementing this, digital forensic through code analysis of file architecture reveals asymmetric encryption: terrain files governing spatial representation are technically protected, while aircraft textures and objects remain modifiable. This material asymmetry demonstrates that neutrality is not incidental, but enforced through design.

The analysis identifies a dual strategy: interface overlays abstract warfare into tactical data (hypermediacy), while selective encryption produces geographically specific yet culturally neutral spaces (immediacy). Together, these strategies enable military simulations to circulate across global markets shaped by conflicting geopolitical investments.

2. Introduction

In an time of unprecedented visual fidelity, digital simulations increasingly mediate how publics understand distant wars—yet their claims to transparent representation mask systematic design choices. This research employs multi-modal forensic methods to investigate how military simulations engineer an appearance of neutrality. Players worldwide engage these simulations as entertainment, navigating photorealistic terrains and operating weapon systems modeled on technical documentation. What hidden architectures shape these experiences? How can investigative approaches reveal how these visual and technical strategies operate?

Digital Combat Simulator (DCS; Eagle Dynamics, n.d.), which markets itself as “the most realistic free-to-play digital battlefield ever,” exemplifies these tensions. A 2024 DCS forum discussion about adding civilian assets for the Afghanistan map revealed critical community engagement with representational boundaries. When users requested market crowds, livestock, and cultural architecture to enable more realistic Rules of Engagement

scenarios, one participant anticipated "we won't get unarmed people for obvious reasons." Another pointed to the paradox: the simulation already permits "laying waste to whole city blocks" and "Afghan villages" — "not exactly the Geneva convention spirit." This exchange reveals systematic patterns: certain forms of violence become acceptable precisely through visual strategies that abstract away human presence. While this analysis focuses on DCS, the platform is treated as an exemplary case through which broader commercial and visual logics of contemporary military simulation can be examined.

Existing scholarship provides crucial frameworks. For example, Dyer-Witheford and de Peuter's (2009) *Games of Empire* establishes that games serve capitalist and imperial interests through labor conditions, distribution networks, and ideological content. Similarly, Bogost (2007) demonstrates how mechanics embed ideology through procedural rhetoric, while Bolter and Grusin (2000) reveal digital media's contradictory logics of hypermediacy and immediacy. However, while Dyer-Witheford and de Peuter analyze political economy broadly, visual strategies as specific market mechanisms remain underexamined. DCS must appeal simultaneously to players in the United States, Russia, China, and Europe—markets with conflicting political investments. Rather than overtly promoting ideology, it achieves its marketability by appearing to promote none through designed visual strategies.

Building on scholars examining the visual politics of war—Bleiker (2018) on visual representation shaping international politics, Butler (2009) on frames determining visible violence and Der Derian (2009) on simulation sanitizing war—this research explores how these insights inform the political economy of game studies. The phenomenon of "commercial neutrality" does not signify the absence of ideology, nor a commitment to realism, rather, it constitutes a market-oriented visual strategy that avoids geopolitical specificity through selective abstraction. Methodologically, while acknowledging artists such as Farocki (2014), who use essay film to critique military simulation, this research experiments with adapting visual methods as systematic analytical tools generating empirical evidence. Rather than functioning as an expressive or interpretive gesture, the essay film is deployed here as a repeatable analytical procedure designed to generate sensory perceptual effects.

This analysis can be understood as a specification and extension of existing frameworks. It develops "commercial neutrality" as a concept that specifies how visual design serves capitalism in contested markets, thereby moving from general political economy to visual political economy. Methodologically, it deploys the essay film as an experimental analytical tool. It integrates insights from political economy (Dyer-Witheford & de Peuter, 2009), visual politics (Bleiker, 2018; Butler, 2009), remediation theory (Bolter & Grusin, 2000), and military geography (Woodward, 2015). Finally, it provides empirical evidence through both technical analysis (terrain encryption) and visual comparison (selective aesthetics).

A note on terminology: Throughout, I use direct language—"war" rather than "conflict," "military" rather than "self-defense"—as a methodological choice to resist the sanitizing discourse under analysis.

The paper proceeds in four sections: theoretical framework (Section 2), analysis of DCS's visual strategies (Section 3), methodology and implications (Section 4), and conclusion.

3.Theoretical Framework

3.1 Double Logic

"Our culture wants both to multiply its media and to erase all traces of mediation: ideally, it wants to erase its media in the very act of multiplying them" (Bolter & Grusin, 2000, p. 5).

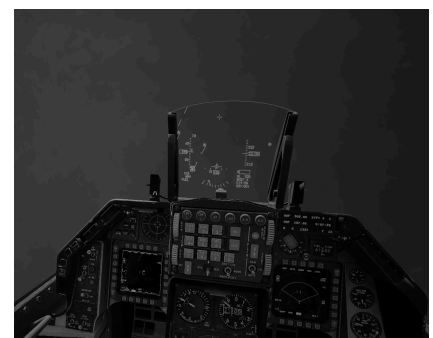
Bolter and Grusin's theory of remediation describes how digital media operate through two contradictory yet coexistent logics: hypermediacy and immediacy. Hypermediacy foregrounds the medium itself—making interfaces, symbols, and technical systems visible and explicit. Immediacy, by contrast, seeks to erase the medium, creating the illusion of transparent, unmediated access to reality. Rather than opposing forces, these logics work together within the same medium to construct specific forms of realism. In video games, this double logic becomes particularly visible: games simultaneously draw attention to their technical systems while claiming to provide immersive, unmediated experiences.

In DCS, hypermediacy operates through pervasive technical interfaces. During flight, the cockpit display (embedded in the model) overlays tactical data—including altitude, speed, weapon status, and navigation waypoints—within the player's view. Instructions are displayed on the upper-right side of the screen. When the aircraft flips, it is shown in slow motion in black and white. The mission editor further exposes the simulation's constructed nature: before gameplay begins, users can adjust weather conditions, set time of day, and configure environmental parameters. Rather than concealing these mediating technologies, DCS foregrounds them. The accumulation of data overlays, system alerts, and configuration options produces a sense of professional authenticity through visible technical complexity.

Simultaneously, DCS pursues immediacy through claims of transparent representation. The simulation bases terrain rendering on satellite geographic data and models aircraft systems with extensive technical documentation. The DCS F-16C manual, spanning 615 pages, meticulously documents aircraft systems and procedures. The simulation employs detailed flight model principles, using aerodynamic calculations to replicate real-world flight dynamics under varying atmospheric conditions (DCS Forum, Documentation). Marketing materials emphasize this fidelity to real-world counterparts, positioning the simulation as



Cockpit view, in-game screenshot, DCS



Aircraft flips in slow motion, in-game screenshot, DCS

providing direct access to actual flight dynamics and combat environments. This pursuit of immediacy operates not by hiding technology but by framing it as purely instrumental—as if the simulation simply channels reality rather than actively constructing it.

These logics function dialectically. Hypermediacy establishes credibility through visible technical systems. Immediacy promises transparent access to reality. Together, they produce a specific form of realism: warfare as both technically complex (requiring system management) and objectively representable (divorced from political or cultural mediation). This double logic shapes not only what players see but how they understand the very nature of simulation.

3.2 Rules as Rhetoric

Video games convey ideology not only through narrative but also through their rules and systems. Ian Bogost's (2007) theory of procedural rhetoric proposes that games persuade through the processes they model and the actions they permit or constrain.

When players engage with game mechanics, they experience and internalize the worldview embedded in those systems (Bogost, 2007). As Jens Seiffert and Howard Nothhaft (2015) observe, "computer games can be procedurally persuasive, mounting an argument through the 'laws,' 'rules,' and available 'procedures' of the world in which the game takes place and in which the player is involved" (p. 706). This persuasion operates through structured interaction rather than explicit statement. Video game Papers, Please (Pope, 2013) exemplifies this mechanism: players assume the role of immigration officers processing documents under time pressure and financial constraints.

The interface appears purely procedural, checking documents against regulations, approve or deny entry. Yet this seemingly neutral bureaucratic process embeds arguments about state power and institutional violence. As players prioritize efficiency over empathy to meet quotas, they experience how systems produce callousness through structural incentives rather than individual malice. The game's rhetoric emerges not from what it says but from what it requires players to do.

Military simulations deploy procedural rhetoric toward different ends. The video game America's Army (U.S. Army, 2002), a government-funded first-person shooter developed as a recruitment tool, models military service as teamwork, discipline, and technical skill. Missions reward coordination and rule-following while penalizing recklessness or deviation from protocol. The game does not explicitly argue for military values—it embeds them in mechanics that define success and failure. Players internalize these values through repeated engagement with systems that make certain behaviors possible, rewarded, or necessary.

In DCS, procedural rhetoric operates along multiple dimensions. Mission outcomes are confined to three possibilities: player death, time expiration, or mission completion. This structure frames warfare as a bounded technical challenge with clear success conditions. The targeting system reinforces this logic: ground targets appear as alphanumeric designators—"Static Cow 1" or "Vehicle Group 3"—accompanied by grid coordinates and tactical data. Buildings become datapoints; convoys become movement vectors. The system rewards precision and efficiency while providing no mechanism to assess civilian presence, cultural significance, or proportionality. Damage assessment generates

numerical readouts—percentage destroyed, objectives updated—abstracting violence into quantifiable outcomes.

These procedural structures embed specific claims: that combat is primarily technical, that success is measured through objective metrics, and that tactical execution can be separated from political or ethical context. Importantly, this procedural rhetoric has a visual dimension. The interfaces, overlays, and data representations that implement these mechanics are not merely functional but rhetorical. They shape how players see the simulated battlefield and what aspects of warfare become visible or invisible.



DCS: mission editor map

3.3 Geography Without Culture

"I feel the making of war resolves itself into very simple issues and the simplest in my view is what is possible and what is not possible."

— Bernard Montgomery (1948)

In this 1948 address to the Royal Geographical Society in London, Field Marshal Bernard Montgomery (as cited in Garver, 2001). He emphasized that in strategic and operational planning, geography fundamentally determines the feasibility and limits of military action. This perspective aligns with Rachel Woodward's (2015) theory in *Military Geography*, which conceptualizes locations as strategic assets and advocates for the quantification and abstraction of space to support strategic decision-making.

The early wargame Kriegsspiel (Wintjes & Pielström, 2018) is considered a prototype of modern military simulation games and exemplifies the military geographic emphasis on tactical terrain reading. The game involves three participants: two generals and one umpire. Generals must read the terrain on a map and write down troop movement orders, which the umpire then executes. Each type of terrain corresponds to different movement speeds, requiring players to make decisions based on the terrain. This example illustrates a core principle of military geography training: the goal is not to memorize place names, but to cultivate the habit of analyzing maps and terrain automatically. In other words, the ability to "read terrain in a de-cultured manner" is not a specific piece of factual knowledge to memorize, but a foundational cognitive skill implicitly embedded in military training exercises.

In the 1960s, the United States Army Command and General Staff College first offered a Military Geography course, and the 1966 publication of *Military Geography* (Peltier & Percy, 1966) became the principal textbook and introductory resource for the discipline. The example of Kriegsspiel demonstrates how such training emphasizes terrain-based decision-making, cultivating analytical thinking rather than rote memorization.



*Army officers playing Kriegsspiel.
Illustration from The Graphic by
Joseph Nash in 17 August 1872*

3.4 From Practice to Market

Originally, military geography served operational needs. With the commercialization of military simulations, DCS adopts these strategies not for operational requirements but to satisfy market demands. As a commercial product, DCS faces a distinct challenge: selling a simulation based on real-world geography to a global audience with divergent geopolitical interests and national allegiances.

This commercial imperative shapes DCS's representational strategy. The simulation must appeal simultaneously to players in the United States, Russia, China, and Europe—markets with conflicting political investments in simulated war. A Syria map that appears partisan toward any faction risks alienating segments of the customer base. In this context, claims of neutrality serve commercial rather than epistemological purposes—a market-driven strategy that maximizes reach by appearing non-partisan.

This strategy operates through generalized visual design. A "universal" visual language removes markers that might signal allegiance or trigger political identification. While DCS claims to "present reality transparently and objectively," in practice it constructs spaces that appear neutral through systematic erasure of ideological markers. Geography is preserved; geopolitics is suppressed.

Commercial neutrality in DCS operates along three dimensions. First, abstraction from cultural and political context: warfare is transformed into decontextualized tactical scenarios, producing geography without geopolitics. Combat occurs in recognizable locations stripped of the cultural and historical meanings that make those locations politically significant. Second, technical and functional framing of combat: engagements are presented as problems to be solved using physics, ballistics, and system mechanics rather than as violent encounters with human consequences. Third, visual design that circumvents ethical concerns: representations stripped of cultural specificity create moral distance, rendering violence as abstract calculation rather than embodied action.

Tensions emerge in the modding community, where players sometimes resist this enforced neutrality. Some customize aircraft with national markings that do not correspond to real-world operations—adding flags or insignia to express political identifications the base simulation avoids. Yet these modifications remain confined to surface-level elements: aircraft skins, objects, missions. DCS's encryption of terrain files ensures that spatial meaning—the core site of commercial neutrality—remains under developer control. Players can mark their allegiances on aircraft but cannot alter the depoliticized spaces through which those aircraft fly.

4. Analysis

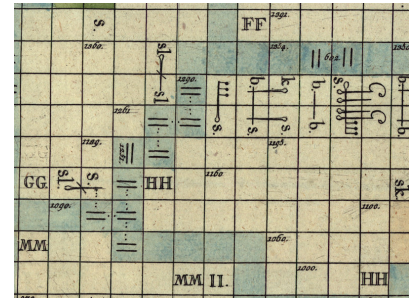
4.1 Hypermediacy, and abstracted Geo

The early wargame Kriegsspiel, developed in 1824 by Prussian artillery officer Georg Heinrich Rudolf Johann von Reisswitz for military training rather than entertainment (Wintjes & Pielström, 2018), is considered a prototype of modern military simulation games and exemplifies the military geographic emphasis on tactical terrain reading.

As a contemporary descendant of war-gaming traditions, DCS inherits this logic while translating board-game abstraction into data-driven simulation. Rivers and forests, now derived from satellite imagery, are already expressions of datafication rather than lived environments. Weather functions similarly: by adjusting numerical parameters, the system can produce rainstorms, sandstorms, or changes in atmospheric conditions. Environmental phenomena thus exist not as ecological systems but as variables within a controllable technical framework.

During gameplay, interface prompts and warning indicators repeatedly interrupt the visual field. Rather than hiding mediating technology, these elements foreground it. The accumulation of technical readouts, alerts, and system messages produces a sense of professional authenticity through technical complexity, exemplifying what Bolter and Grusin describe as hypermediacy. The player is constantly reminded that they are managing systems rather than inhabiting an unmediated world.

This technical framing is further reinforced by what the system chooses *not* to simulate. Environmental elements respond only to aircraft mechanics, not to violence or atmospheric force. Trees do not fall during storms, animals do not flee from explosions, and civilians do not escape collapsing buildings. These visual absences are not gameplay limitations but rhetorical devices. They visually assert that combat is a matter of system management rather than killing, framing warfare as problem-solving. In this sense, the abstraction of geography functions as the visual dimension of procedural rhetoric.



Historical wargame, known as Hellwig's Game, Versuch eines aufs Schachspiel gebaueten taktischen Spiels (Attempt at a tactical game based on chess) published in 1780

4.2 Immediacy, and cultural unreadability

Provided as DCS's default free environment, the Caucasus map renders buildings with geometric accuracy but textures them with moss and vines, suggesting abandonment. In other maps, buildings appear newer, and recognizable landmarks are geometrically accurate and based on real locations. However, textures are simplified, replacing ornamental details with flat color surfaces. Even in prominent sites such as the plaza of the Great Mosque of Herat, the DCS environment remains entirely empty. By contrast, images of the same locations on Google Maps frequently show crowds and everyday activity.

Similar patterns appear in other maps: the Europatum tower in the Cold War Germany map and the Eiffel Tower in *DCS: Normandy 2.0* are faithfully modeled, yet the surrounding streets remain devoid of human presence. DCS's promotional claim for Normandy 2.0—"over 100,000 buildings, towns, and cities that you can explore"—demonstrates that the inclusion of landmarks is not avoided. Architectural specificity is a selling point.

This selective representation is reinforced at the technical level. DCS terrain files are encrypted, preventing players from modifying geography. At the same time, the system allows extensive modding through open access to object-import pipelines. Players can introduce new aircraft skins, objects, and visual elements, even though the entire codebase could technically be encrypted. This asymmetry reveals that neutrality in DCS is not a technical constraint but a design decision.

What is removed is not geography but cultural readability—the markers and traces that would identify who inhabits these spaces, what social practices occur within them, or what cultural values shape their architecture. These environments are sufficiently specific to satisfy claims of realism, yet sufficiently generic to suppress ethical and political response. Rendering techniques and texture choices sever architecture from lived meaning, erasing traces of daily life to produce moral distance. This strategy enables immediacy: the illusion of transparent access to reality, while quietly neutralizing its social implications.

Military geography emphasizes that both physical and cultural characteristics shape military operations. Yet in DCS, only the physical dimension is rendered operative. Cultural presence is visually excluded.

(1)



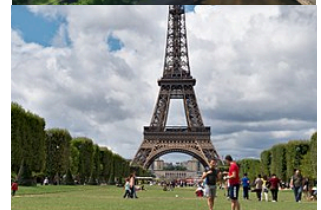
(2)



(3)



(4)



(1) Herat Central Blue Mosque, in-game screenshot, DCS map Afghanistan

(2) Herat Central Blue Mosque, Wikimedia Commons

(3) Eiffel Tower, in-game screenshot, DCS map Normandy 2.0.

(4) Eiffel Tower, Wikimedia Commons

4.3 Possible reads on static 01

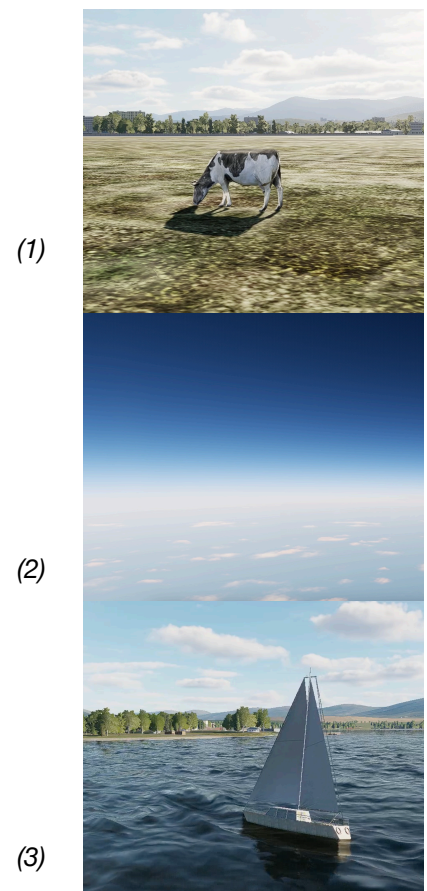
The preceding sections demonstrate how DCS simultaneously deploys two seemingly contradictory visual strategies. Hypermediacy, through technical interfaces and data overlays, frames warfare as system management. Immediacy, through cultural erasure, produces spaces that appear neutral and transparent. How do these logics operate together?

An interview with a long-term DCS player reveals the selectivity underlying these strategies (personal communication, April, 2025). When asked about his experience with the simulation, he made a striking observation: "The map is functional to me—I use it for planning routes, identifying targets. But the clouds are beautiful" This split perception—functional terrain versus aesthetic sky—points to a broader pattern in DCS's visual investments. Beyond clouds, DCS populates its environments with numerous non-functional decorative elements. Sailboats drift in coastal harbors. Cows graze in rural fields. Lighthouses stand on shorelines. The moon and stars are rendered dynamic lighting. Waves break realistically on beaches. None of these elements serve tactical purposes—they exist purely for atmospheric immersion, demonstrating that DCS invests substantial rendering resources in aesthetic detail.

Yet these decorative elements share a crucial characteristic: cultural and political neutrality. Cows signify "rural" without specifying which culture's agriculture. Sailboats suggest "coastal activity" without indicating any particular nation's maritime tradition. Celestial objects carry no geopolitical affiliation. Lighthouses mark navigation without political boundaries. These objects can be beautiful because they are semiotically inert—they communicate generic environmental qualities rather than specific cultural or political meanings.

This selective aesthetic investment becomes revealing when contrasted with what DCS systematically excludes from its decorative vocabulary. Market stalls displaying culturally specific goods remain absent—they would be too identifiable. Architectural ornamentation that might indicate religious buildings is stripped away—too politically charged. Street vendors, café terraces, and daily social life disappear—too humanizing of conflict zones. Graffiti, posters, or contemporary cultural signifiers are nowhere to be found—too contextual and potentially controversial.

The technical capacity exists to render all these elements. If DCS can animate grazing livestock and simulate realistic



(1) Static cow 01, in-game screenshot

(2) Skyline, in-game screenshot

(3) Static boat, in-game screenshot

wave physics, it could certainly render market awnings or architectural details. The absence is not a limitation but a choice. DCS can afford to be beautiful—clouds can be stunning, animals can populate fields, boats can sail—but only in domains where beauty carries no geopolitical stakes. Buildings and urban environments, however, must remain generic, because architectural and cultural specificity might identify the very communities over which players conduct their missions.

This pattern reveals that DCS's visual strategy is not uniform realism but strategic authenticity. The simulation selectively deploys high visual fidelity across different spatial and cultural domains. In politically neutral territories—sky, nature, generic maritime or agricultural elements—DCS pursues aesthetic richness and immersive detail. In potentially controversial domains—architecture, urban culture, religious or ethnic markers—it enforces neutrality through systematic erasure.

The coexistence of hypermediacy and immediacy thus serves a unified commercial logic. Technical interfaces (hypermediacy) establish professional credibility and simulation authenticity. Cultural unreadability (a specific form of immediacy) maintains market neutrality by avoiding partisan identification. Selective aesthetics—knowing precisely where to invest in beauty and where to enforce abstraction—enables DCS to appear both realistic and apolitical, satisfying simultaneously the demands for immersive experience and global marketability.

4. Method & Conclusion

4.1 Methodology:

The visual strategies analyzed in this research—selective aesthetics, cultural erasure, engineered neutrality, these are difficult to fully capture through textual description alone. Reading about perceptual ambiguity differs fundamentally from experiencing it. This research therefore employs an essay film not as illustration but as analytical method, generating embodied evidence that textual analysis cannot produce.

After introduction, the film rapidly alternates between DCS screenshots and real-world footage of comparable environments. The editing rhythm varies strategically: sometimes cutting every 2-3 seconds, creating rapid perceptual shifts; other times holding longer, allowing sustained observation before transition. This variable pacing deliberately produces moments of uncertainty where viewers repeatedly confront the question: "Which am I watching, simulation or reality?"

The knowledge generated emerges through viewer experience. Audiences attempt to distinguish simulated from real environments and fail to do so consistently. This perceptual failure itself constitutes evidence: if observers cannot reliably differentiate DCS's rendering from photographic documentation, the simulation has achieved visual indistinguishability, but not through comprehensive realism but through strategic cultural erasure. Buildings appear real because they lack the markers that would identify them as culturally specific. The absence of detail creates the illusion of objective representation. This method produces knowledge that is sensory and affective rather than propositional. The sensation of uncertainty, the moment of misrecognition, the inability to locate oneself between simulation and reality, these cannot be adequately verbalized and they must be felt. A potential critique might frame perceptual evidence as insufficiently rigorous—mere subjective response rather than objective analysis. However, this critique presumes that

visual representation itself achieves stable objectivity. Yet visual strategies themselves remain subject to critical scrutiny. A forum discussion titled 'Towns have too many trees - too green' (DCS World Forums, 2025) demonstrates this instability: a player suggested that tree density in the Germany map exceeds reality, attaching real flight footage as evidence. Despite DCS's pursuit of neutral objectivity, players continuously project real-world knowledge onto the simulation, contesting its representational authority through external verification. If visual representation itself is not truly objective, then perceptual instability is not a flaw in the evidence but a means of exposing visual authority. Methods generating perceptual uncertainty engage the very site where visual politics operates—where claims to neutrality are made and negotiated.

4.2 Conclusion

This research began with a question raised by DCS community forums: what makes a war simulation's representation "appropriate"? The debate over whether to include civilian assets revealed an underlying tension—certain forms of violence remain representable as long as their human cost stays invisible.

Analysis demonstrates that DCS constructs commercial neutrality through deliberate visual architecture. Hypermediacy frames warfare as system management through technical interfaces. Immediacy operates through cultural erasure, producing spaces that appear geographically specific yet culturally generic. Selective aesthetics—beautiful clouds, grazing livestock, empty streets—reveals that visual fidelity is strategically deployed according to commercial and political calculus rather than uniform pursuit of realism.

The simulation's claim to transparent, objective representation masks ideological construction. Geography is preserved; cultural meaning is systematically removed. This is not technical limitation but design decision, serving commercial imperatives to appear neutral while maximizing global marketability.

War simulations do not simply represent war—they construct specific ways of seeing it. The question is not whether simulations are neutral but whose interests their engineered neutrality serves and what forms of understanding it forecloses. In claiming objectivity, DCS produces a politically consequential form of visibility: war as abstract, technologized space emptied of the communities over which it is conducted. This analysis has implications for understanding simulation and visual culture more broadly. The relationship between visual design and political economy becomes clear. Simulations are not merely technical achievements but commercial products whose visual choices reflect market positioning and geopolitical calculation. DCS's selective aesthetics is investing in beautiful clouds while stripping architecture of cultural markers, it serves commercial rather than representational logic. Neutrality is not absence of politics but a specific form of political-economic management.

This pattern extends beyond DCS. When technical products require global markets, neutrality becomes a commercial imperative realized through specific visual strategies. Digital environments negotiate cultural specificity against market reach. In each case, engineered neutrality serves particular interests while marginalizing alternative perspectives.

Cultural erasure shapes how participants understand war zones, training a decontextualized gaze that normalizes abstracting away cultural complexity and creates

moral distance from violence. Yet players are not passive recipients. Forum debates reveal critical awareness, and modding communities push against imposed boundaries, suggesting that while visual strategies shape perception, they do not determine it absolutely. Future research might examine how other military simulations handle cultural representation, investigate long-term effects on players' geopolitical understanding, or explore counter-hegemonic visual strategies that resist commercial neutralization and center marginalized perspectives.

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